

SUBHAM KUMAR

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Subham400 — Subham Kumar — Portfolio

EDUCATION

Institute of Engineering and Management, Kolkata

B.Tech in Electrical and Electronics Engineering; CGPA: 7.69 / 10.0

2026

Salt Lake Sector-V

EXPERIENCE

Internship at School Of Mechatronics & Robotics, IEST, Shibpur

June-September, 2025

- Implemented Kalman filtering algorithms on real-time flight data to improve drone odometry accuracy and stability.
- Handled sensor calibration and drone flight operations for autonomous system workflows.

Internship at ElectroMeter Corporation (EMC)

January-February, 2025

- Designed custom PCB boards for IoT applications and integrated them with hardware.
- Developed and deployed a full-stack web interface to monitor real-time IoT device telemetry.

PROJECTS

Local SIEM Lab | VirtualBox · Ubuntu VM · Security Monitoring (2026)

Personal Project · 2026

- Built a local Security Information and Event Management (SIEM) lab environment on a host machine using an Ubuntu VM in VirtualBox.
- Configured log aggregation, telemetry parsing, and custom detection rules to monitor system and network events locally.

Local Retrieval-Augmented Generation (RAG) System | Python · PyTorch · Transformers

Personal Project · 2026 · [local-rag](#)

- Built an end-to-end local RAG pipeline featuring PDF parsing, sentence-level chunking, and semantic vector retrieval.
- Ran local inference using Gemma-2B (4-bit NF4 quantization) with chain-of-thought context injection for zero-API-cost answering.

AI Voice Keyboard — Android IME | Python · Kotlin · FastAPI · CUDA

Personal Project · 2026 · [whisper-voice-keyboard](#)

- Developed an Android speech-to-text keyboard leveraging faster-whisper and Gemma for real-time transcript cleanup.
- Built a CUDA-accelerated FastAPI backend communicating with a native Kotlin IME client via local Wi-Fi with Silero VAD.

Custom Flight-Controller-Based Quadcopter (ESP32 & STM32 | Embedded Systems & Robotics)

College Project · 2025-2026

- Developed a quadcopter flight controller handling real-time sensor processing and state estimation for GPS-denied environments.
- Integrated magnetic sensors and data-logging pipelines to enable aerial magnetic mapping and geophysical analysis.

Full-Stack Course Selling Website | [Live](#) · Full-Stack Web Development

Client Project · 2025

- Built and deployed a production course-selling website with secure authentication, payment gateway(Razorpay) integration, and responsive UI.

TECHNICAL SKILLS

AI/ML: RAG, LLMs, Quantization (4-bit NF4/GGUF), Semantic Search, Embeddings, PyTorch, Transformers, OpenCV, TensorFlow

Languages & Web: Python, Kotlin, Java (Basics), HTML, CSS, JavaScript, SQL, PHP, FastAPI, React, Next.js

Tools/Platforms: VirtualBox, Ubuntu/Linux, Git, GitHub, CUDA, llama.cpp, Android Studio, MATLAB-Simulink, EasyEDA

Domains: AI/ML, Cybersecurity, Computer Vision, IoT, Full Stack Web Dev, Embedded Systems, Android Development

CERTIFICATIONS

[Machine Learning Foundations](#) | [Data Analytics](#) | [Excel-Data Analysis](#) | [Data Visualization](#) | [SQL](#) | [Java Algorithms](#)
[Cyber Security Fundamentals](#) | [Ethical Hacking](#)